Name_____ Date_____ Period_____

LESSON 5-4 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

Area of Regular Polygons

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a regular polygon by finding the area of one triangle and multiplying.

REMEMBER

- **Regular polygon**: equal sides, equal angles.
- Lines from the center to each vertex make *n* **congruent triangles**. Each one's height is the **apothem**.
- **A = n × (½ × side × apothem)**. Use ≈ if rounded.

WORKED EXAMPLE

Regular hexagon: side 8 cm, apothem ≈ 6.9 cm.

Step 1 A hexagon makes 6 congruent triangles.

Step 2 One triangle: ½ × 8 × 6.9 = 27.6

Step 3 Six triangles: 6 × 27.6 = 165.6

Answer **≈ 165.6 cm²**

WORD BANK

regular polygon – all sides equal, all angles equal

apothem – center to the middle of a side

congruent – same size and same shape

≈ – about

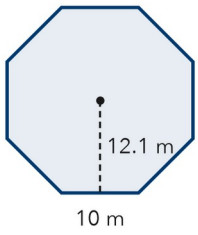
PRACTICE

show your work · label units

1. Find the area of the regular octagon. The apothem is rounded.

Step 1 Find one triangle: ½ × 10 × 12.1.

Step 2 An octagon has 8 triangles. Multiply by 8.



Show your work.

Answer_____ m²

2. Complete the table. Each apothem is rounded, so mark each area with ≈.

Step 1 Perimeter = number of sides × side.

Step 2 Area ≈ ½ × perimeter × apothem.

Regular polygon	Side	Apothem	Perimeter	Area
Hexagon	4 cm	3.5 cm		
Pentagon	10 ft	6.9 ft		

3. Each regular polygon has perimeter P and rounded apothem a. Find each area. Mark it with ≈.

a. pentagon: P = 30 cm, a ≈ 4.1 cm ____ cm²

b. hexagon: P = 60 ft, a ≈ 8.7 ft ____ ft²

4. Gazebo Floor. A gazebo floor is a regular **hexagon**. Each side is 6 ft. The apothem is about 5.2 ft. Tile costs $3 for each square foot. About how much does the tile cost?

Step 1 Find the area: 6 × (½ × 6 × 5.2).

Step 2 Multiply the area by $3.

Show your work.

Answer_____

5. Find the Error. Andre says a regular pentagon with sides of 8 cm and an apothem of 5.5 cm has area ½ × 8 × 5.5 = 22 cm². Explain his mistake and find the area.

Frame *Andre found only ___ triangle. A pentagon has ___ triangles, so the area is ___.*

_____

_____

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___

Name_____ Date_____ Period_____

LESSON 5-4 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Composite Figures

Maryland Standard 6.G.A.1 · Learning target – I can find the area of a composite figure by adding or subtracting the areas of simpler shapes.

REMEMBER

- A **composite figure** is made of simpler shapes.
- **Add** the areas of joined shapes. **Subtract** a piece that is cut out.
- Use the sides you know to find missing sides.

WORKED EXAMPLE

House front: a 10 ft × 6 ft rectangle with a triangular roof on top. The roof's base is 10 ft and it is 4 ft tall.

Step 1 Rectangle: 10 × 6 = 60

Step 2 Triangle: ½ × 10 × 4 = 20

Step 3 The shapes are joined, so add: 60 + 20 = 80

Answer **80 ft²**

WORD BANK

composite figure – a shape made of simpler shapes

add – put areas together

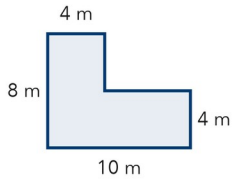
subtract – take away a cut-out area

right angle – a square corner

PRACTICE

show your work · label units

1. Find the area of the L-shaped figure. All corners are right angles.



Step 1 Cut the L into two rectangles.

Step 2 Find each rectangle's area.

Step 3 Add the two areas.

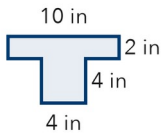
Show your work.

Answer_____ m²

2. Write **add** or **subtract** to tell how you would find each area.

a. a wall, not counting its window: _____ **b.** a bedroom plus its closet: _____

3. Find the area of the T-shaped figure. All corners are right angles.



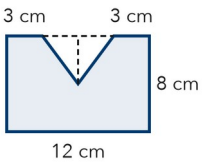
Step 1 Cut the T into a top bar and a stem.

Step 2 Find each rectangle's area.

Step 3 Add the two areas.

Answer_____ in²

4. A triangle notch 4 cm deep is cut from the top of a rectangle. Find the area that is left.



Step 1 Find the rectangle: 12 × 8.

Step 2 Notch base: 12 − 3 − 3. Find the triangle.

Step 3 Subtract the triangle.

Answer_____ cm²

5. **Backyard.** A yard is a **rectangle**, 20 ft by 12 ft. A square garden is in one corner. Each side of the garden is 5 ft. The garden is **not** paved. Find the paved area.

Step 1 Find the yard area.

Step 2 Subtract the garden area.

Answer_____ ft²

TEST PRACTICE

choose the best answer

6. A rectangle 9 m by 6 m has a square 2 m by 2 m cut out. What area is left?

A 50 m² **B** 58 m² **C** 54 m² **D** 13 m²

7. A 10 in by 4 in rectangle is joined to a triangle with base 4 in and height 3 in. What is the total area?

A 34 in² **B** 40 in² **C** 46 in² **D** 52 in²

CHECK YOURSELF	□ I can do this on my own	□ I need a little help	□ I have a question about #___